MILITARY SPECIFICATION

BALLISTICS COMPUTER, SUBASSEMBLY: 8635370 (COMPUTER UNIT)

1. SCOPE

1.1 This specification covers one type of electrical-mechanical assembly known as Ballistics Computer, Subassembly: 8635370, hereinafter referenced as "Computer Unit" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–F–13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.
MIL–P–14232 —Parts, Equipment and Tools for Ordnance Materiel, Packaging of.
MIL–I–45208 —Inspection System Requirements.
MIL–I–45607 —Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.
MIL–STD–202 —Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8635370 —Ballistics Computer, Subassembly: 8635370 (Computer Unit).
IEL 8635370 —List of Inspection Documents for Ballistics Computer, Subassembly: 8635370.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The computer unit shall be manufactured in accordance with Drawing F8635370 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.

3.3 General specification. The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL–F–13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.4 Environmental.

3.4.1 *Operating temperatures.* The computer unit shall meet the requirements of 3.5 to 3.6.4 inclusive while exposed and thermally stabilized at ambient temperatures of —40 and +150°F. Upon return to standard ambient temperature the computer unit shall meet the requirements of 3.4.2.

3.4.2 *Vibration.* The computer unit less shock mounts shall be examined for loose parts, defects or damage and shall meet the requirements of 3.5 to 3.6.6 inclusive after being vibrated in a vertical plane for five minutes ±15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second (cps).

3.4.3 *Water resistance (sealing).* The interior of the computer unit shall show no evidence of water penetration or other damage after being subjected to a simulated rainfall for a time duration of 15 minutes ±15 seconds.

3.5 Electrical.

3.5.1 *Continuity.* The continuity of the computer unit shall conform to Drawing F10516313.

3.5.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the computer unit are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts plus twice the rated voltage at a frequency of 60 cps for a period of not more than 30 seconds nor less than five seconds.

3.5.3 *Insulation resistance.* All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 10 megohms with 500 volts direct current (dc) applied.

3.6 Performance.

3.6.1 *Input voltages.* The computer unit shall meet the requirements of 3.6.2 thru 3.6.4 when excited with 18 to 30 vdc between pin K of J606 and pin r of J607, pin r to receive the positive connection, with 115 ±5%, 400 cps ac rms volts between pins F and G of J606.

3.6.2 *Ammunition selection.* Switch S601 shall step to and indicate selection of operator selected ammunition positions when J602 and J603 are connected to switch and indicator devices electrically appropriate as a medium to control ammunition selection. The maximum time required for 3 steps (ammunition 1 to 4, 2 to 1, etc.) shall not exceed 3 seconds.

3.6.3 *Effective full charge count.* The effective full charge dial shall advance 1 count on APDS–M392–A2, 4 counts on APDS–M392–A1, 1/10 count on HEP–M393, and 2/5 count on HEAT–M356 for each actuation of a test switch connected to J601 which is electrically appropriate as a medium to control actuation of the EFC circuit. Dial backlash to be removed before each test.

3.6.4 *Scale factor.* The scale factor adjustments of Drawing F8635370, sheets 8 and 9, shall be confirmed. R601 thru R608 and R613, shall be within ±.0019 of the ratios indicated. R618 shall be within ±0.01 of the ratio indicated, R609 thru R612, R620 and R621 shall be within ±.0010 of the ratios indicated. R614 and R617 shall be within 5 ohms and 2 ohms respectively of the values indicated.

3.6.5 *Elevation amplifier output.* The maximum input signal required at J4 of the elevation summing network to cause ampli-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.

fier saturation (maximum current flow through the center tap of 6.0 ohms center tapped resistive load connected between pins L and M of J605 with the center tap supplied from pin J of J605) shall not exceed 0.080 volts rms at computing phase with gain control R20 full counterclockwise. J2, J3, J5, J6, J7 and J8 of the elevation summing network shall be grounded, the dc supply voltage of 3.6.1 shall be set to 24 volts, the ac supply shall be the same as 3.6.1, and the test temperature shall be standard ambient.

3.6.6 *Deflection amplifier output.* The maximum input signal required at J4 of the deflection summing network to cause amplifier saturation (maximum current flow through the center tap of a 106 ohm center tapped resistive load connected between pins C and F of J604 with the center tap supplied from pin D of J604) shall not exceed 0.080 volts rms at computing phase with gain control R20 full counterclockwise. J2, J3, J5, J6, J7, and J8 of the deflection summing network shall be grounded, the dc supply voltage of 3.6.1 shall be set to 24 volts, the ac supply shall be the same as 3.6.1, and the test temperature shall be standard ambient.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or purchase order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and the Supplementary Quality Assurance Provisions listed in the List of Inspection Documents, IEL 8635370. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105 and the inspection level and sampling plan specified in table I titled "Classification of Defects". Examinations and tests for packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement and to reject individual non-conforming items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.

TABLE I. *Classification of defects*

Use Level II of Table I with Sampling Plan Table II-A of Standard MIL-STD-105

	Requirement	Test procedure
Critical: None.		
Major: AQL 1.5%		
101. Physical failure, post vibration.	3.4.2	4.5.1.2
102. Continuity	3.5.1	4.5.2
103. Ammunition selection	3.6.2	4.5.5.2
104. Effective full charge count	3.6.3	4.5.5.3
105. Scale factor	3.6.4	4.5.5.4
106. Elevation amplifier output.	3.6.5	4.5.5.5
107. Deflection amplifier output.	3.6.6	4.5.5.6
Minor: None.		

Note. The above tests shall be performed at standard ambient temperature (60° to 90°F.).

4.2.3 *Disposition of nonconforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests.

4.3.1 *General.* One unit shall be selected at random from each one hundred produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	Requirement	Test procedure
301. Materials	3.1	As per spec or dwg
302. Fabrication	3.2	Visual—per dwg
303. General specification	3.3	Visual—MIL-F-13926

4.3.2 *Environmental and electrical.* Three units shall be selected at random as control samples from each 50 produced, or from each month's production, whichever occurs first. The control samples shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

Control test	Requirement	Test procedure
304. Operating temperatures.	3.4.1	4.5.1.1
305. Water resistance	3.4.3	4.5.1.3
306. Dielectric strength	3.5.2	4.5.3
307. Insulation resistance	3.5.3	4.5.4

4.3.3 *Control test failure.* Should any one item of the control test sample fail to meet the specified requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirements of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspection to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2 *Inspection equipment accuracy requirements.* The devices required in the test methods and procedures herein, to measure the electrical values specified in Section 3,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.

shall provide readings to the accuracy specified below:

Item Number	Description
1. Ratio Standard	Ratio range from .00000 to 1.00000 accuracy to ±.01%. Input impedance at 400 cps, 500,000 ohms minimum. Series impedance — 10 ohms maximum. Phase shift 0.01° maximum.
2. Oscilloscope	Minimum sensitivity of 20 millivolts, full scale, vertical channel, External synchronization. input impedance — 1 megohm ±10%.
3. A.C. Power Supply	115 volts rms, 400 cps ±20 cps, 5% maximum harmonic distortion.
4. D.C. Power Supply	18–30 volts, maximum ripple—1.0 volt peak to peak, 15 ampere output.
5. Ammo Selector Test Switch	In accordance with fig. 1. Minimum switch contact rating — 10 amps, 24 VDC.
6. E.F.C. Test Switch	In accordance with fig. 2, with provisions for holding switch in battery position a minimum of 2 seconds per actuation.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 *Operating temperatures.* This test shall be conducted in conformance with the temperatures specified in 3.4.1. The test equipment shall be in accordance with the "Test Facilities" requirements specified in Specification MIL–F–13926. While thermally stabilized at the specified temperatures, the computer unit shall be subjected to the tests specified in 4.5.5 to 4.5.5.4 inclusive to determine conformance with the requirements referenced therein.

4.5.1.2 *Vibration.* To determine conformance to 3.4.2, the computer unit shall be mounted in its normal position using a suitable mounting apparatus to attach it to a vibrating machine or fixture. Test equipment shall be in accordance with the "Test Facilities" portion of Specification MIL–F–13926 and shall be capable of providing vibration at the amplitude and frequency for the time specified in 3.4.2. After vibration, the computer unit shall be given a visual—tactile examination for damage and loose parts. The computer unit shall then be subjected to the tests specified in 4.5.2 to 4.5.5.6 inclusive to determine conformance with the respective requirements referenced therein.

4.5.1.3 *Water resistance.* To determine conformance to 3.4.3, this test shall be conducted in accordance with Procedure I of Specification MIL–F–13926 except for the time duration specified in 3.4.3. The computer unit shall show no evidence of water penetration as a result of this test when visually examined.

4.5.2 *Continuity.* The computer unit shall be tested, using standard measuring equipment, to demonstrate continuity of the circuits in accordance with Schematic Drawing F10516313 to determine compliance with 3.5.1.

4.5.3 *Dielectric strength.* The computer unit shall be tested in accordance with Method 301 of Standard MIL–STD–202 and shall meet the requirements specified in 3.5.2.

4.5.4 *Insulation resistance.* Immediately after satisfactorily withstanding the test for dielectric strength, the unit shall be given the insulation resistance test in accordance with Method 302, test condition B, of Standard MIL–STD–202, and shall meet the requirements specified in 3.5.3.

4.5.5 *Performance.* The computer unit shall be subjected to performance tests, using test setups and testing devices as specified in 4.4.2. Equipment and unit being tested shall be connected in accordance with Electrical Schematic F10516313, Wiring Diagram F10516360, and figures 1 and 2 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.

4.5.5.1 *Input voltage tests.* To determine conformance to 3.6.1, make the DC and AC hookups as specified. For each of the tests 4.5.5.2 thru 4.5.5.4, make the first check at 24 VDC input, then check at 18V and again at 30V.

4.5.5.2 *Ammunition selection.* To determine conformance to 3.6.2, observe and time the switching action of S601 as the Ammo Selector Test Switch (fig. 1) is actuated at random throughout the various possible combinations of selecting ammunition. The computer unit shall be as specified.

4.5.5.3 *Effective full charge count.* To determine conformance to 3.6.3, trip the E.F.C. test switch 10 times with APDS–A1 ammo in the circuit. Observe E.F.C. dial for 40 count movement. Select APDS–A2 ammo and trip E.F.C. test switch 60 times. Observe E.F.C. dial for 60 count movement. Select HEP ammo and trip E.F.C. test switch 400 times. Observe E.F.C. dial for 40 count movement. Select HEAT ammo and trip E.F.C. test switch 150 times. Observe the E.F.C. dial for 60 count movement. For each ammo selected, the E.F.C. counter shall perform as specified.

Note. Allow a minimum of 2 seconds to elapse for each E.F.C. test switch actuation.

4.5.5.4 *Scale factor.* To determine conformance to 3.6.4, connect the ratio standard, power suppliers, and oscilloscope as specified on Dwg. F8635370. Proceed to check out each adjustment as specified on the computer unit drawing. The various resistors shall be within the tolerances specified in 3.6.4.

4.5.5.5 *Elevation amplifier output.* To determine conformance to 3.6.5, set the DC supply voltage at 24V and the AC supply voltage at 115V 400 cps. Ground J2, J3, J5, J6, J7 and J8 of elevation summing network. Turn R20 full counterclockwise. With the specified load resistors connected between pins L, M, and J of J605, read the input signal at J4 of the elevation summing network required for amplifier saturation. The input signal under these conditions shall not exceed 0.080 volts RMS.

4.5.5.6 *Deflection amplifier output.* To determine conformance to 3.6.6, set the DC supply voltage at 24V and the AC supply voltage at 115V—400 cps. Ground J2, J3, J5, J6, J7 and J8 of the deflection summing network. Turn R20 full counterclockwise. With specified load resistors connected between pins C, F, and D of J604, read the input signal at J4 of the deflection summing network required for amplifier saturation. The input signal under these conditions shall not exceed 0.080 volts RMS.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL–P–14232 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. The computer unit is primarily intended as a functional component of the electronic analog Ballistics Computer: M16. The unit is designed to perform functions of ammunition selection switching, summation of input and followup signal, amplification of servo error signals and power distribution.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing (see 5.1).
(c) Applicable Packaging Data Sheet Number (see 5.1).
(d) Applicable stock number.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. 1220–A165Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.

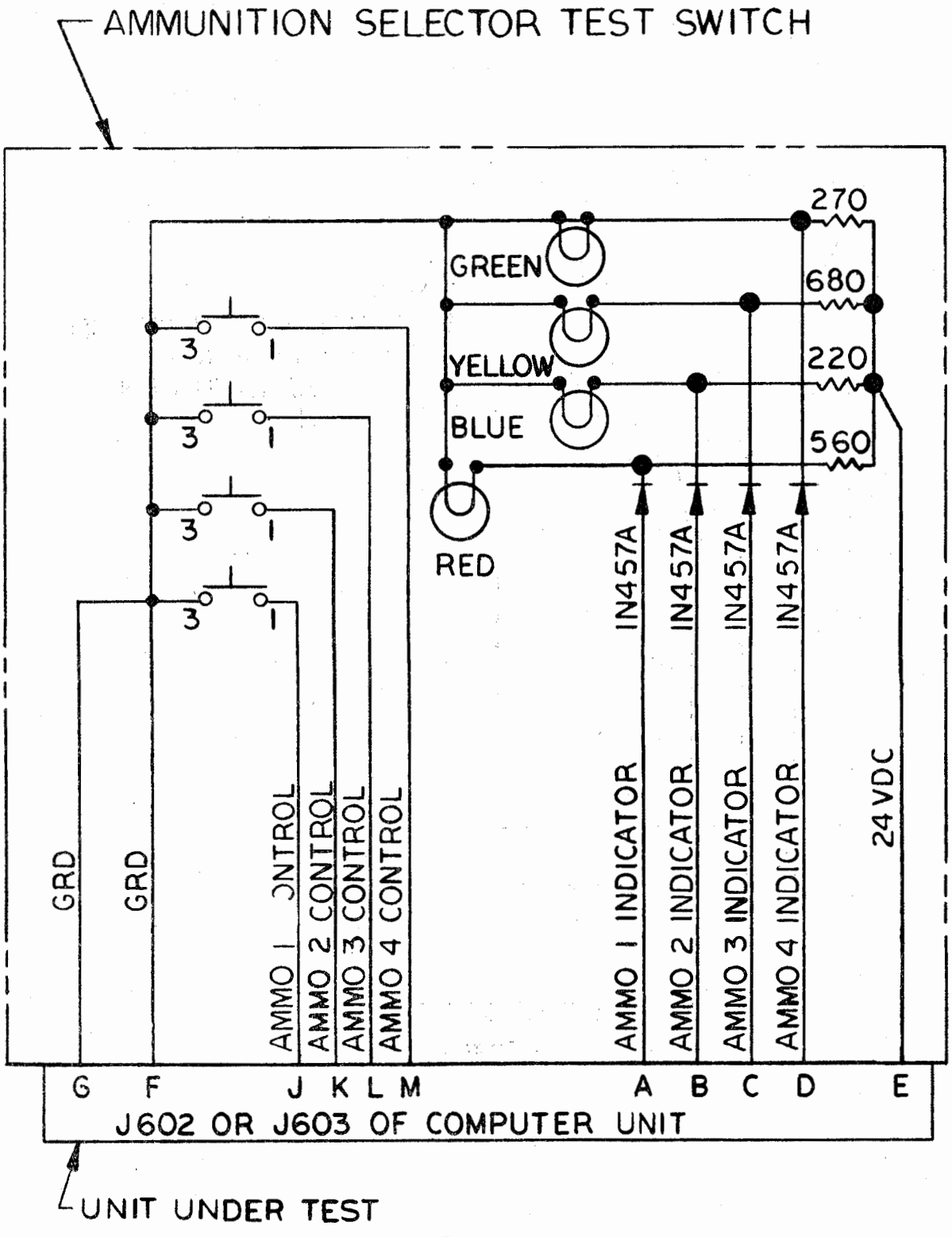


FIGURE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.

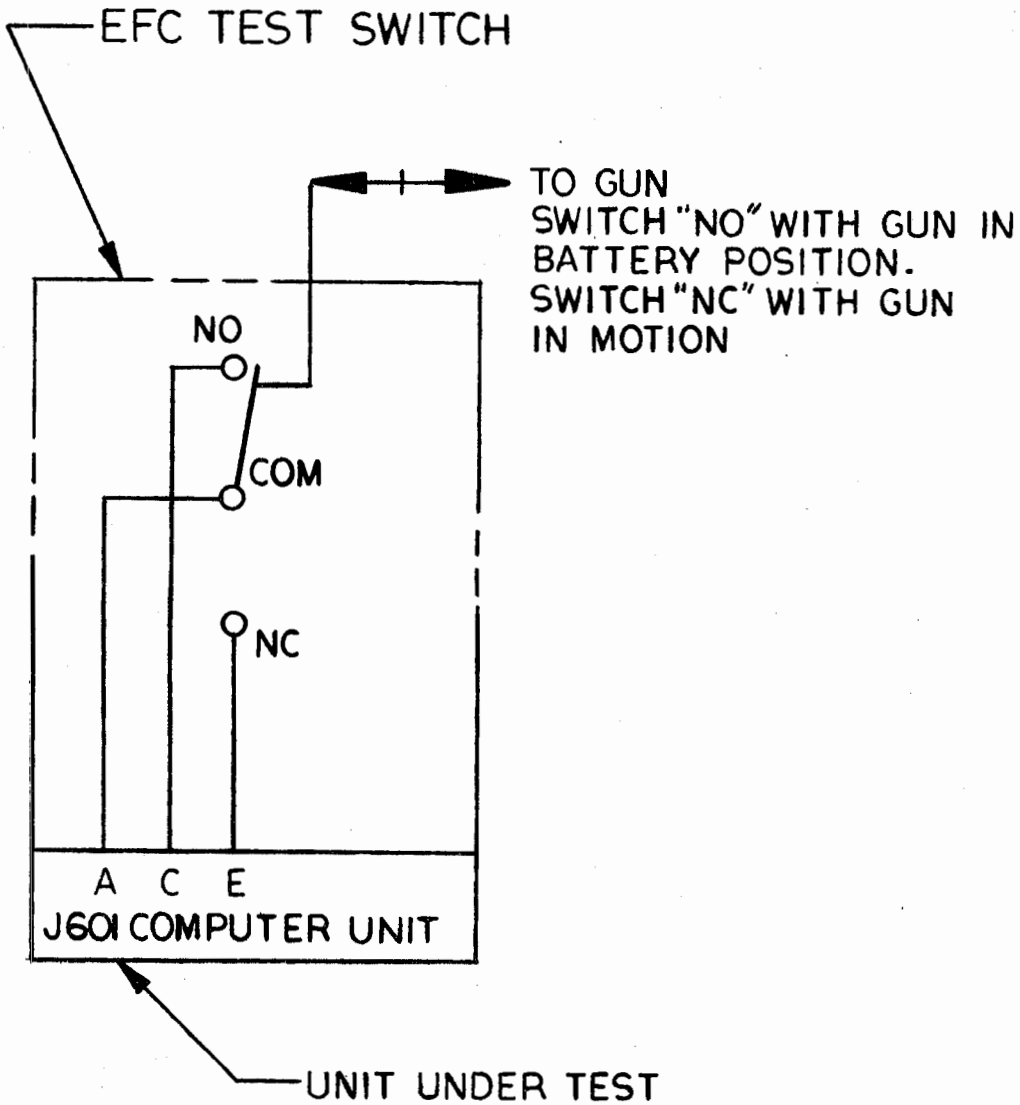


FIGURE 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:11Z
Check the source to verify that this is the current version before use.